

Simulation-Based Inference

Resampling methods

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The concept of *resampling* refers to the repeated sampling from the same original, finite dataset or (more generally) from any finite set of samples.

Common types of resampling:

- Bootstrapping
- Permutation tests
- Cross-validation
- Jackknife

Main idea: Instead of relying on knowing the true (analytic) sampling distribution (or normal approximations thereof), bootstrapping uses the observed data samples as a replacement for the population.

Advantage over the simulation-based testing approach: We do not need to fix/know the true parameter θ , but can work with the estimate $\hat{\theta}$ as the “approximate truth” around which to construct a sampling distribution.

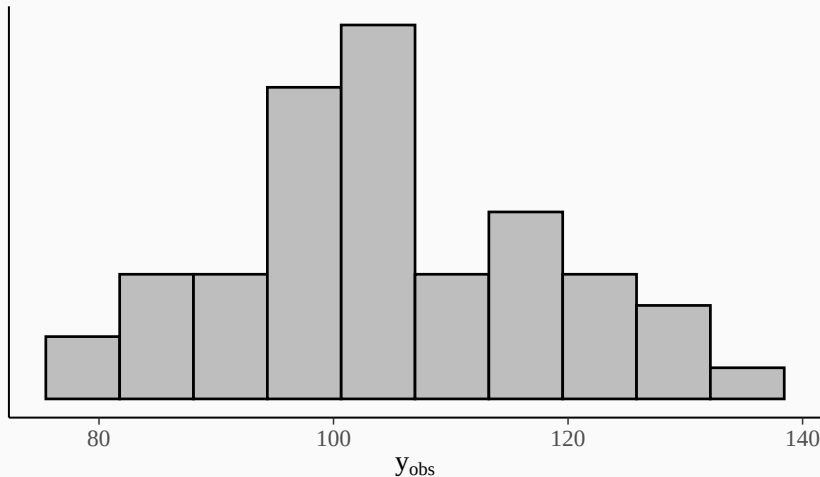
In other words, the true model M_1 underlying the observed data y does not have to be known but is implicitly given via resampling from y .

Bootstrap: Algorithm

- Start with a dataset $\{y_i\}_{i=1}^N$, which we assume to have exchangeable observations y_i . Compute the estimate $\hat{\theta}$ of θ from $\{y_i\}_{i=1}^N$.
- Choose the number B of bootstrap replications (datasets).
- For each $b = 1, \dots, B$ randomly draw N values with replacement from the original data $\{y_i\}_{i=1}^N$ to obtain $\{y_i^{(b)}\}_{i=1}^N$
- The result is a set of (bootstrapped) datasets: $\{\{y_i^{(b)}\}_{i=1}^N\}_{b=1}^B$
- Compute estimate $\hat{\theta}^{(b)}$ of θ for each of the B bootstrap datasets.
- Further process the set of bootstrap estimates $\{\hat{\theta}^{(b)}\}_{b=1}^B$.

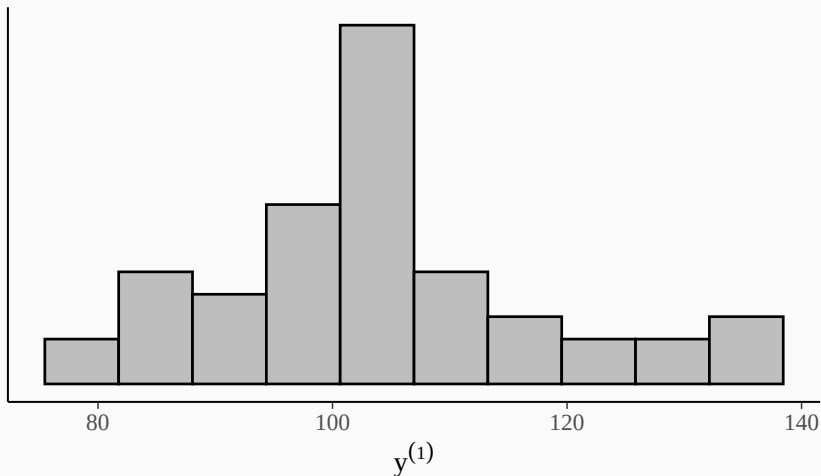
Example: Mean intelligence of students

We assume the same data and context as earlier.



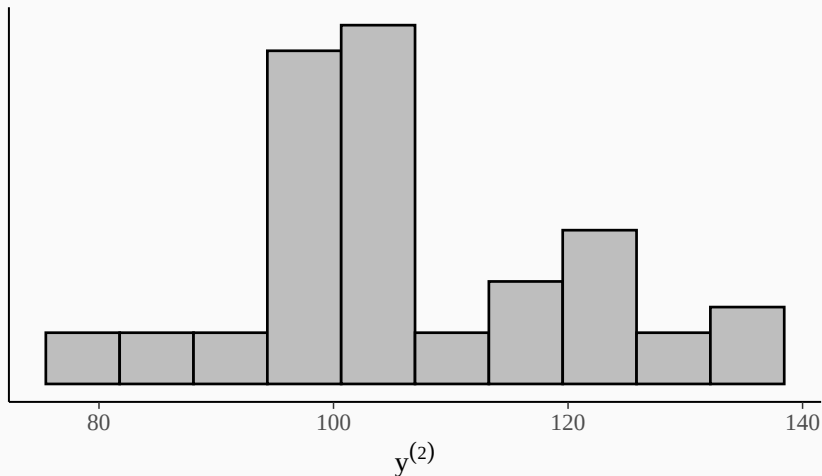
Empirical mean $\hat{\mu} = \bar{y} = 104.52$, variance $\hat{\sigma}^2 = 186.33$.

First Bootstrap replication



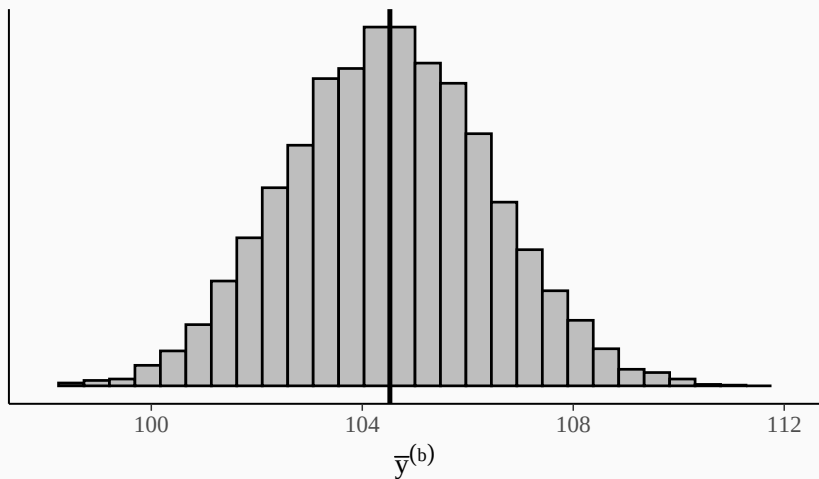
Bootstrap mean $\hat{\mu}^{(1)} = 103.83$, variance $(\hat{\sigma}^2)^{(1)} = 198.63$.

Second Bootstrap replication



Bootstrap mean $\hat{\mu}^{(2)} = 106.13$, variance $(\hat{\sigma}^2)^{(2)} = 190.62$.

Bootstrap sampling distribution for $B = 10000$



Bootstrap for bias correction

Many estimates $\hat{\theta}$ have non-zero bias, at least for finite N :

$$\text{Bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta \neq 0$$

The bootstrap estimate of the $\text{Bias}(\hat{\theta})$ is

$$\text{Bias}^{(B)}(\hat{\theta}) = \bar{\theta}^{(B)} - \hat{\theta}_{\text{obs}}$$

where $\bar{\theta}^{(B)} = \sum_{b=1}^B \hat{\theta}^{(b)} / B$ is the mean of the bootstrap estimates.

A bias-corrected estimate of θ can thus be computed as

$$\hat{\theta}_{\text{obs}}^{(B)} = \hat{\theta}_{\text{obs}} - \text{Bias}^{(B)}(\hat{\theta}).$$

Bootstrap for standard error estimation

The standard errors (SE) of an estimator $\hat{\theta}$ of θ is defined as the (true) standard deviation $\text{SD}(\hat{\theta}(y))$ assuming $y \sim p(y \mid \theta)$.

For S simulated datasets $y^{(s)} \sim p(y \mid \theta)$ producing estimates $\hat{\theta}^{(s)} := \hat{\theta}(y^{(s)})$, we can approximate the SE by

$$\text{SE}^{(S)}(\hat{\theta}) = \sqrt{\frac{1}{S} \sum_{s=1}^S (\hat{\theta}^{(s)} - \bar{\theta}^{(S)})^2}$$

For B bootstrapped datasets $y^{(b)}$, producing estimates $\hat{\theta}^{(b)} := \hat{\theta}(y^{(b)})$, we can approximate the SE by

$$\text{SE}^{(B)}(\hat{\theta}) = \sqrt{\frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \bar{\theta}^{(B)})^2}$$

Bootstrap for confidence intervals

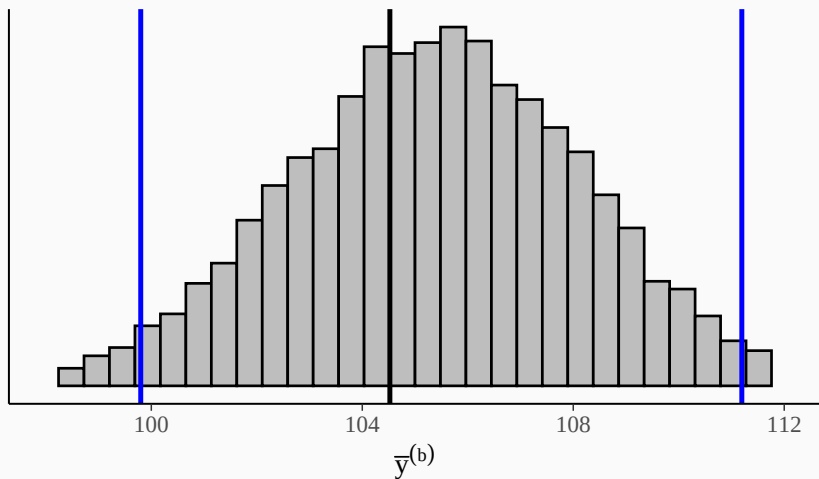
One way to obtain $1 - \alpha$ confidence intervals with bootstrap, are *Bootstrap percentile intervals* (BPIs).

Compute the empirical quantiles $Q_{\alpha/2}^{(B)}$ and $Q_{1-\alpha/2}^{(B)}$ from the set $\{\hat{\theta}^{(b)}\}_{b=1}^B$.

Then, $[Q_{\alpha/2}^{(B)}, Q_{1-\alpha/2}^{(B)}]$ is a bootstrap confidence interval for the quantity of interest θ

BPIs are not the only way to define bootstrap confidence intervals as we will see.

Bootstrap for confidence intervals of the mean



The empirical variance is usually defined as:

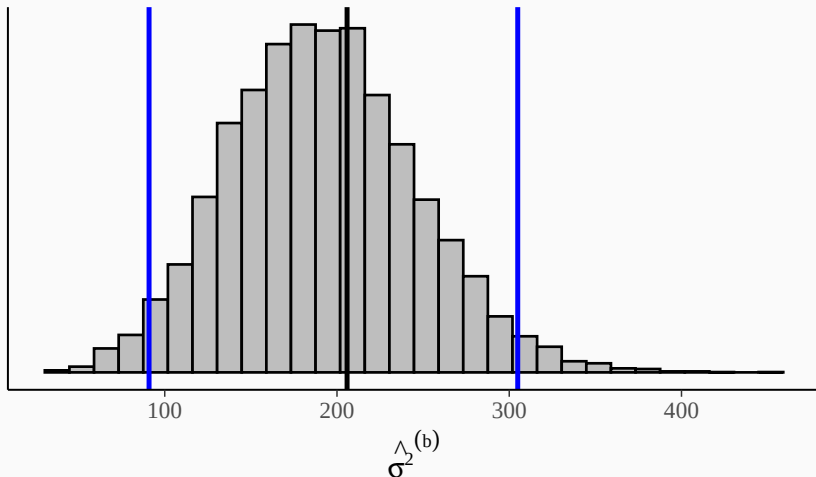
$$\hat{\sigma}^2(y) = \frac{1}{(N-1)} \sum_{i=1}^N (y_i - \bar{y})^2$$

If the data are normally distributed, $\hat{\sigma}^2(y)$ is an unbiased and consistent estimator of the true skewness σ^2 .

Note: Under normality, the Variance estimator $\hat{\sigma}^2$ follows a scaled χ^2 distribution.

Bootstrap for confidence intervals of the variance

Use a subset of $N = 15$ observations to illustrate stronger asymmetry.



Question: Do you think the BPI is a valid CI for θ here?

Centering bootstrap samples

Let us consider the true sampling distribution $p(\hat{\theta} - \theta)$ (with $\hat{\theta}$ being random), which we seek to approximate.

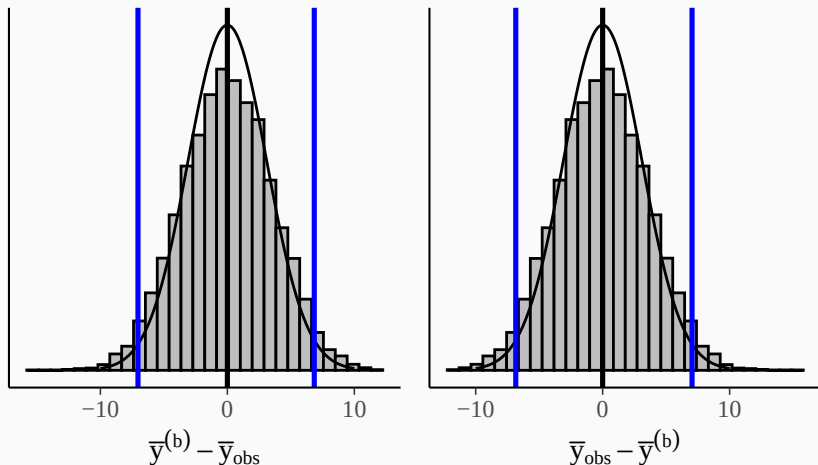
We consider two alternative *centered* bootstrap distributions as candidates to approximate $p(\hat{\theta} - \theta)$.

- The distribution $p(\hat{\theta}^{(b)} - \hat{\theta}_{\text{obs}})$
- The distribution $p(\hat{\theta}_{\text{obs}} - \hat{\theta}^{(b)})$

Question: Which of these two candidates you think is better?

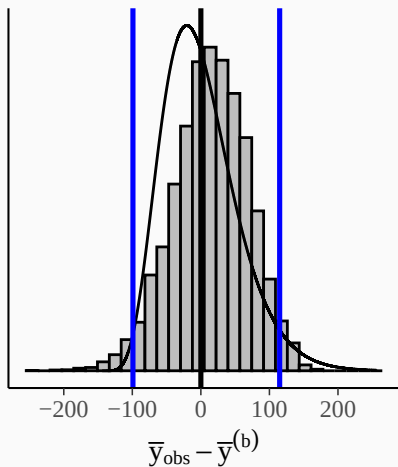
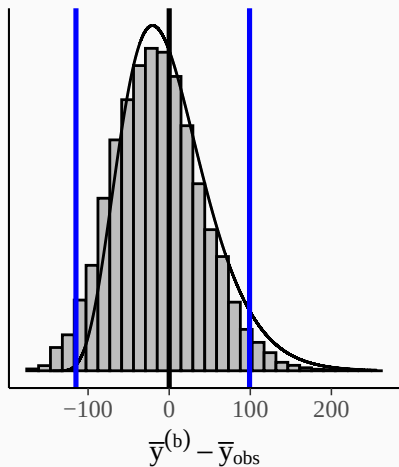
Question: How does that help us with constructing CIs?

Centered bootstrap distributions for the mean



Density lines indicate the true sampling distribution of $\bar{y} - \mu$.

Centered bootstrap distributions for the variance



Centered bootstrap percentile intervals

For *symmetric* true sampling distributions:

- Both centered bootstrap distributions are equal in the limit of $B \rightarrow \infty$:

$$p(\hat{\theta}^{(b)} - \hat{\theta}_{\text{obs}}) = p(\hat{\theta}_{\text{obs}} - \hat{\theta}^{(b)})$$

- Both approximate the true centered sampling distribution $p(\hat{\theta} - \theta)$.

For *asymmetric* true sampling distributions:

- Only $p(\hat{\theta}^{(b)} - \hat{\theta}_{\text{obs}})$ approximates $p(\hat{\theta} - \theta)$
- However, only $p(\hat{\theta}_{\text{obs}} - \hat{\theta}^{(b)})$ produces valid confidence intervals, called *centered bootstrap confidence intervals* (CBCIs):

$$\text{CBCI}_{\alpha}^{(B)}(\theta) = [2\hat{\theta}_{\text{obs}} - Q_{1-\alpha/2}^{(B)}, 2\hat{\theta}_{\text{obs}} - Q_{\alpha/2}^{(B)}]$$

Implication: (Non-centered) BPIs are valid CIs for θ only if the (true) sampling distribution is *symmetric*.

Limitations of bootstrap

- For small datasets (low N), the bootstrap approximation of the population may be inappropriate.
- Heavy tailed distributions or outliers may be challenging for bootstrap.
- Parts of the population not covered by the data won't be covered by the bootstrap.
- For high accuracy $B \geq 1000$ or even $B \geq 10000$ may be required, which becomes computationally challenging if computing the estimator $\hat{\theta}$ is slow.
- Bootstrap for dependent data structures (e.g., clusters, time-series, spatial data) requires special resampling algorithm to correctly account for the dependencies.

Permutation tests

Permutation tests enable statistical testing of hypotheses relating to the comparison of two or more groups.

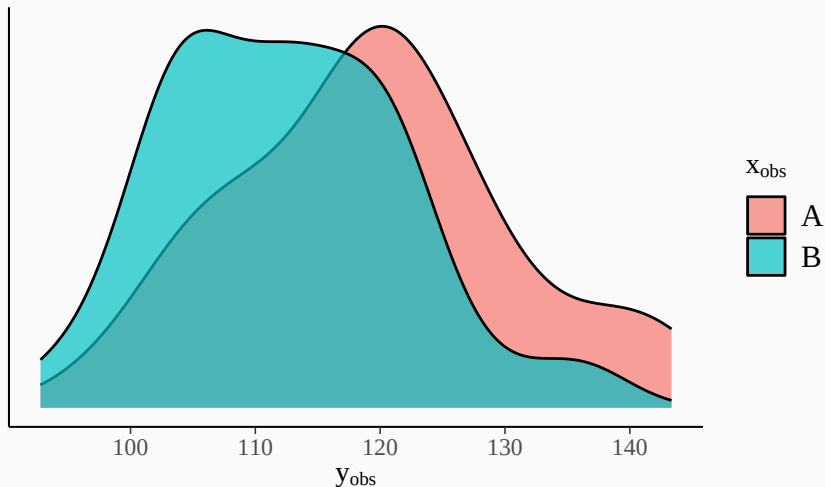
Two group case: Observed dataset $D_{\text{obs}} = \{y_i, x_i\}_{i=1}^N$ with responses y_i and group labels $x_i = 0, 1$.

Procedure:

- Compute the estimate $\hat{\theta}_{\text{obs}} := \hat{\theta}(D_{\text{obs}})$ based on the observed data
- Choose the number of permutations P .
- For each $p = 1, \dots, P$:
 - Randomly permute the group labels thus creating a new dataset $D^{(p)} = \{y_i, x_i[p]\}_{i=1}^N$, where $x_i[p]$ is the new (permuted) label of observation i .
 - Compute the estimate $\hat{\theta}^{(p)} := \hat{\theta}(D^{(p)})$ for the p th permuted dataset
- Store all such estimates in a set $\{\hat{\theta}^{(p)}\}_{p=1}^P$.
- Compare $\hat{\theta}_{\text{obs}}$ with $\{\hat{\theta}^{(p)}\}_{p=1}^P$.

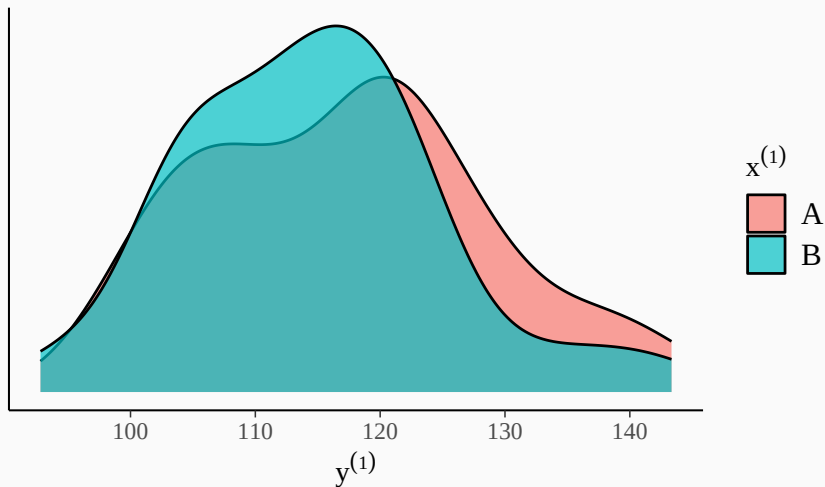
Example: Comparison of two groups

Hypothetical comparison of students from University A vs. B



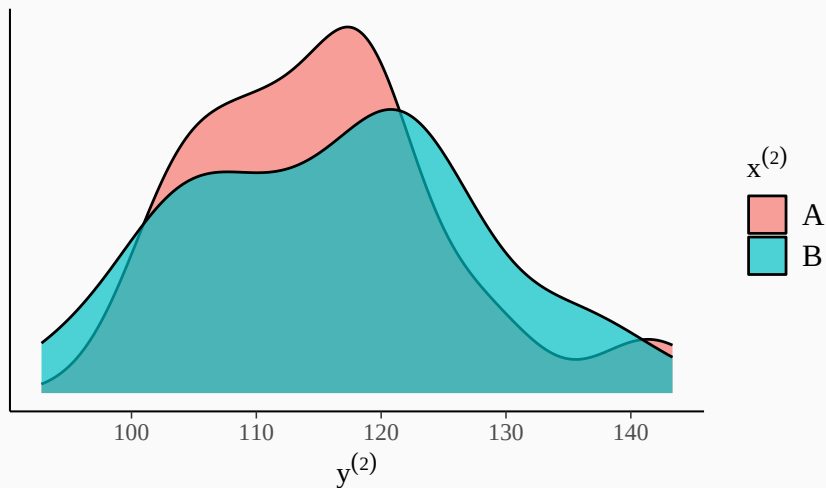
Mean difference $\delta_{\text{obs}} = 7.01$.

First permuted dataset



Mean difference $\hat{\delta}^{(1)} = 2.69$.

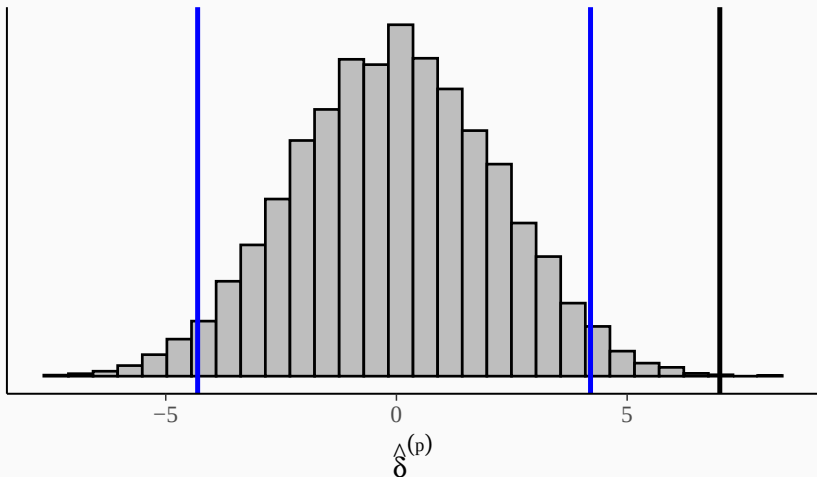
Second permuted dataset



Mean difference $\hat{\delta}^{(2)} = -0.83$.

Permutation-based distribution of the mean difference

Permutation-based distribution of the mean difference $\hat{\delta}^{(p)} = \hat{\mu}_A^{(p)} - \hat{\mu}_B^{(p)}$



Black line: Observed mean difference. Blue lines: 95% CI.

- Only simple hypotheses concerning the comparison of groups can be easily tested.
- Permutation-based approximations of the sampling distribution may be inappropriate for small datasets.
- Computationally intensive due to the repeated evaluation of the estimator.
- Permutation-tests for non-exchangeable data are harder to derive.

Frequentist predictions

Assume a model M with likelihood $p(y \mid \theta)$.

Inference: Given observed data y_{obs} obtain an estimate $\hat{\theta}_{\text{obs}}$ of θ .

Prediction (sampling version): Simulated S (model-implied) datasets $y^{(s)} \sim p(y \mid \hat{\theta}_{\text{obs}})$.

Evaluation (sampling version): Compare observed and (model-implied) simulated data via an appropriate metric (loss function):

$$\bar{L} = \frac{1}{S} \sum_{s=1}^S L(y^{(s)}, y_{\text{obs}})$$

Example (squared loss):

$$L(y^{(s)}, y_{\text{obs}}) = \sum_{i=1}^N (y_i^{(s)} - y_{\text{obs},i})^2$$

In-sample vs. out-of-sample predictions

In-sample predictions:

- How close are the model's predictions to the data it was estimated on?
- Problem: High danger of overfitting

Out-of-sample predictions:

- How close are the model's predictions to new data?
- Ideally balances under- and overfitting
- Problem: How do we evaluate predictions on new data without actual new data?

Steps in cross-validation (with validation = test data):

- (1) Split the data into two subsets: training data and test data
- (2) Fit the model on the training data
- (3) Evaluate the predictions on the test data
- (4) Repeat (1) to (3) with multiple data splits
- (5) Summarize the results of all splits

Types of cross-validation (selection)

Leave-one-out cross-validation (LOO-CV):

- Each time, use one observation as the test data and the remaining $N - 1$ observations as training data.
- Ideally, evaluate all N possible LOO-folds.

K-fold cross-validation (K-fold-CV):

- Each time, use N/K observations as the test data and the remaining $N - N/K$ observations as training data.
- Each observation is in the test set only once.
- Evaluate all such K folds.

Types of cross-validation (selection; continued)

Leave-one-group-out cross-validation (LGO-CV):

- For grouped (clustered) data: Use all observations of a single group as test data and the rest as training data.
- Ideally, evaluate all K possible folds, where K is the number of groups.
- Also applies to spatial data (leave-one-location out).

Leave-future-out cross-validation (LFO-CV):

- For time-series data: Train on the past, test on the future.
- Usually, training (and test data) vary in size across folds.

“Leave-neighborhood-out cross-validation” (my name):

- For spatial data: Each fold, choose one location for testing.
- Train on all non-neighbors of that location.
- Ideally, evaluate for all K possible folds.

Apply cross-validation (usually LOO-CV) but for inference rather than prediction.

Shares the same goals as bootstrapping but uses methods of cross-validation.

Concretely: Leave out one observation at a time from the dataset y and recalculate $\hat{\theta}$ each time. For a dataset with N observations, we obtain N different “leave-one-out” estimates $\hat{\theta}^{(i)} := \hat{\theta}(y_{-i})$.

Jackknife for bias correction

Again, let $\hat{\theta}$ have non-zero bias, at least for finite N :

$$\text{Bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta \neq 0$$

The Jackknife estimate of the $\text{Bias}(\hat{\theta})$ is

$$\text{Bias}^{(J)}(\hat{\theta}) = (N - 1) \left(\bar{\theta}^{(J)} - \hat{\theta}_{\text{obs}} \right)$$

where $\bar{\theta}^{(J)} = \sum_{i=1}^N \hat{\theta}^{(i)} / N$ is the mean of the jackknife estimates.

Question: Why do we need to multiply by the factor $(N - 1)$?

A bias-corrected estimate of θ can thus be computed as

$$\hat{\theta}_{\text{obs}}^{(J)} = \hat{\theta}_{\text{obs}} - \text{Bias}^{(J)}(\hat{\theta}).$$

Reminder: The SE of an estimator $\hat{\theta}$ of θ is defined as the (true) standard deviation $\text{SD}(\hat{\theta}(y))$ assuming $y \sim p(y \mid \theta)$.

For N Jackknife estimates $\hat{\theta}^{(i)}$, we can approximate the SE by

$$\text{SE}^{(J)}(\hat{\theta}) = \sqrt{\frac{N-1}{N} \sum_{i=1}^N (\hat{\theta}^{(i)} - \bar{\theta}^{(J)})^2}$$

Instead of $\bar{\theta}^{(J)}$ we may also use $\hat{\theta}_{\text{obs}}$ as the “reference estimate.”

Question: When do we expect the two resulting SE estimates to differ?

Instead of resampling the original dataset, we can also fully simulate datasets *from the model* fitted to the original dataset.

Reminder: The logistic regression model

$$y_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = \text{logistic}\left(\sum_{k=0}^K b_k x_{ik}\right)$$

has no closed form estimators of the coefficients b_i . Standard (iterative) estimators, such as MLE, have a sampling distribution that is normal only asymptotically (for large enough N).

Example: Predicting heart disease with logistic regression

Predicting the probability of having a heart disease by age and sex. For illustration, we use only a small data subset of $N = 20$ observations.

First few rows of the data:

disease	age	sex
1	0.7136023	male
0	-1.4964070	female
1	1.7081065	male
0	-1.1649056	male
0	0.4926014	male
1	0.9346033	female

Fitting a logistic regression model

We fit the following model M to the small heart disease data:

$$\text{disease}_i \sim \text{Bernoulli}(\pi_i) \quad \pi_i = \text{logistic}(b_0 + b_1 z_{\text{age},i} + b_2 \text{sex})$$

```
fit_obs <- glm(disease ~ zage + sex, data=heart_small, family=binomial())  
summary(fit_obs)
```

Call:

```
glm(formula = disease ~ zage + sex, family = binomial(), data = heart_small)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.6672	1.0709	-0.623	0.5332
zage	1.0459	0.5978	1.749	0.0802 .
sexmale	1.1226	1.2269	0.915	0.3602

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Simulating from a logistic regression model

Goal: Obtain a confidence interval for b_1 based on model predictive simulations.

The logistic model M_{obs} obtain from fitting to the original data is

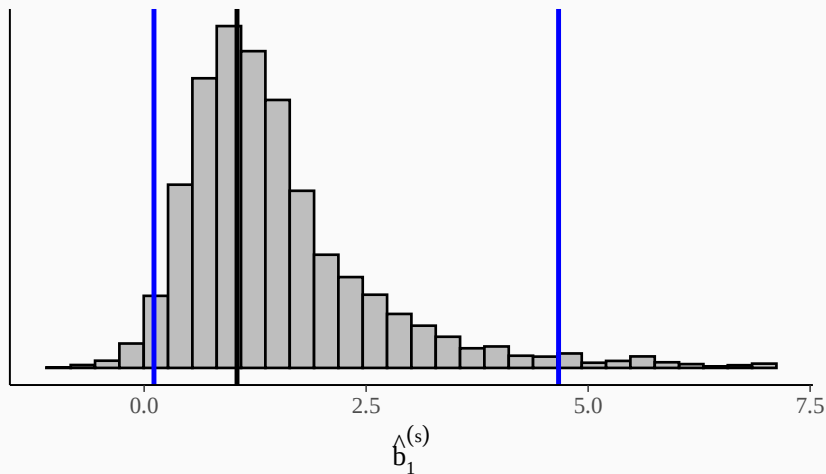
$$\text{disease}_i \sim \text{Bernoulli}(\hat{\pi}_i) \quad \hat{\pi}_i = \text{logistic}(\hat{b}_0 + \hat{b}_1 z_{\text{age},i} + \hat{b}_2 \text{sex})$$

With all parameters b_0 , b_1 and b_2 being set to their estimates.

We then fit the model M to the simulated responses with the predictor values taken from the real data.

Question: What practical differences are there between bootstrapping and the model-based simulation procedure?

Sampling distribution of the age regression coefficient



Black: Estimate from the original data. Blue: 95% BPI.